Macronutrient intake, plasma large neutral amino acids and mood during weight-reducing diets.

Influence of diet composition on mood during weight-reducing diets was studied in healthy young women of normal weight. A broad range of macronutrient intake was achieved by means of divergent dietary instructions for the composition of a 1,000 kcal per day diet adhered to for six weeks. Global mood during the last three weeks of the diet was significantly better in the "vegetarian" than in the "mixed" diet group. During this time a significant correlation was observed between relative carbohydrate intake and global mood ($r = -0.74$; p less than 0.01) and between the ratio of plasma tryptophan to other large neutral amino acids (a predictor of tryptophan flow into brain) and global mood ($r = -0.52$; p less than 0.05). Results suggest that group differences are related to differences in carbohydrate intake. It is hypothesized that impairment of central serotonergic function due to reduced tryptophan availability can prompt mood deterioration in situations of relatively low carbohydrate intake.